Unit 5
Resistors

Unit 5 Resistors

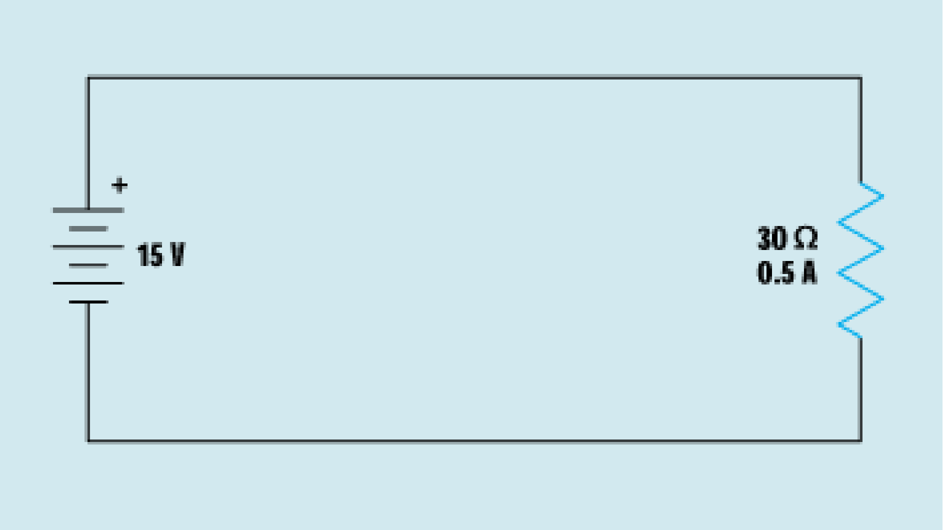
Objectives:

- List the major types of fixed resistors.
- Determine the resistance of a resistor using the color code.
- Discuss how exceeding its power rating can cause damage to a resistor.
- Discuss the use of a variable resistor as a potentiometer.

Unit 5 Resistors

- Resistors are commonly used to perform two functions in a circuit.
- The first use is to **limit the flow** of current in a circuit.

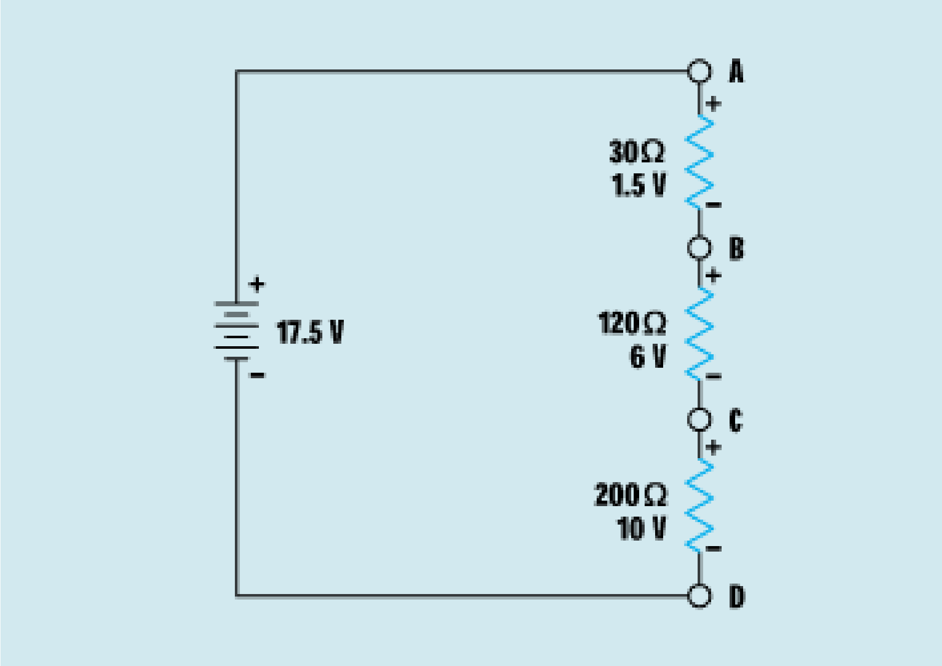
$$I = E / R$$
$$I = 15\ V / 30\ \Omega$$
$$I = 0.5\ A$$



Unit 5 Resistors

- Resistors are commonly used to perform two functions in a circuit.
- The second use is to produce a **voltage divider**.

A to B = 1.5 V
A to C = 7.5 V
A to D = 17.5 V
B to C = 6 V
B to D = 16 V
C to D = 10 V



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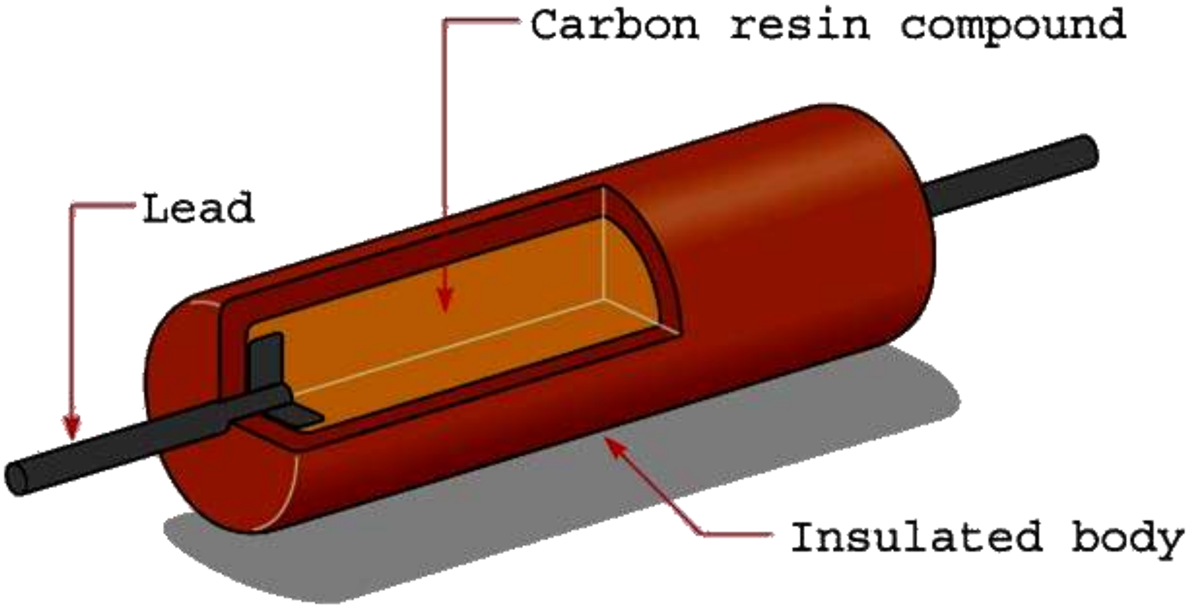
Fixed resistors have only one ohmic value, which cannot be changed or adjusted. One type of fixed resistor is the composition carbon resistor.



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Carbon resistors are very popular for most applications because they are inexpensive and readily available in standard sizes and wattages.

½ Watt 1 Watt 2 Watt

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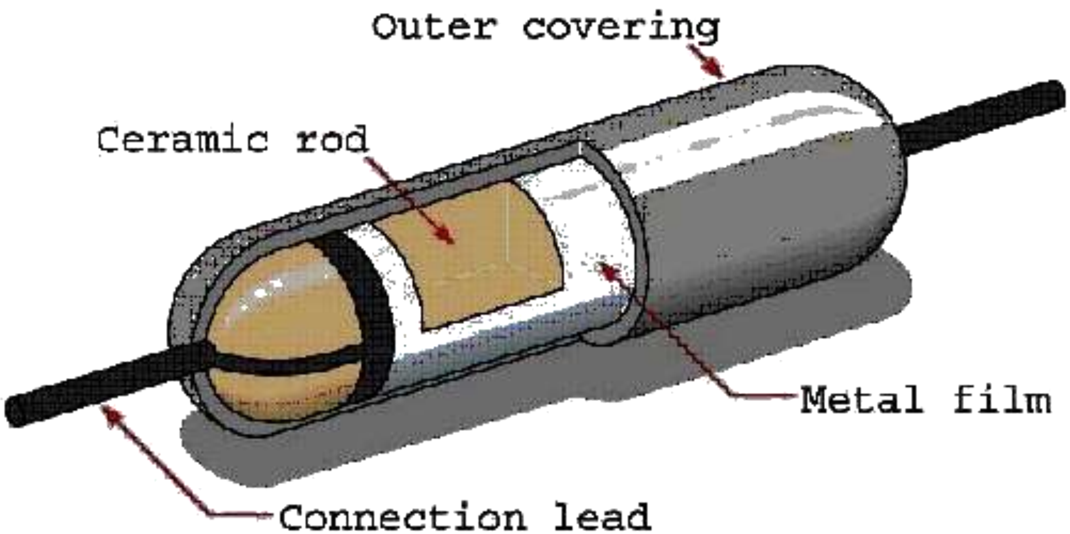
Metal film resistors are another type of fixed resistor. These resistors are superior to carbon resistors because their ohmic value does not change with age and they have improved tolerance.



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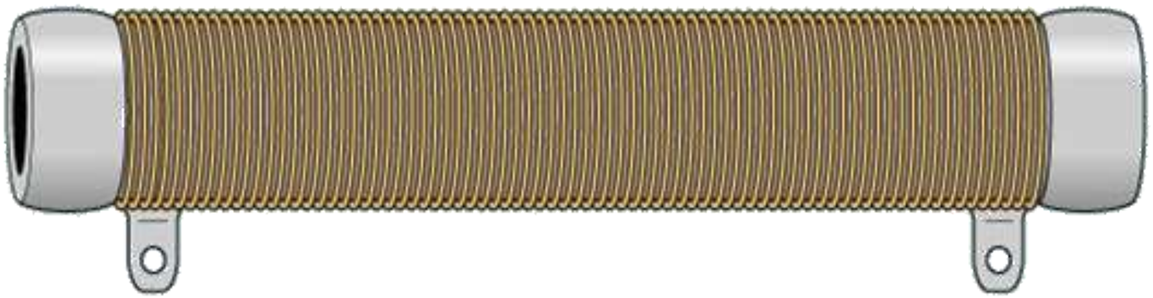
Wire-wound resistors are fixed resistors that are made by winding a piece of resistive wire around a ceramic core. These are used when a high power rating is required.



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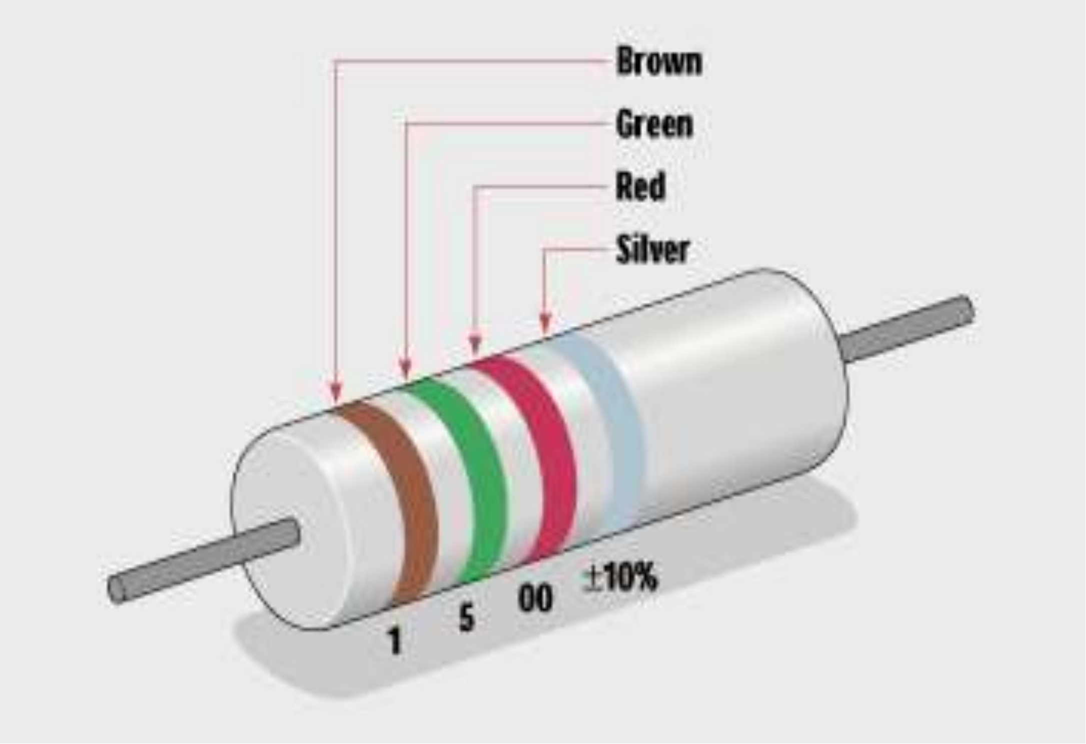
The **resistor color code** can be used to determine the resistor's ohmic value and tolerance.



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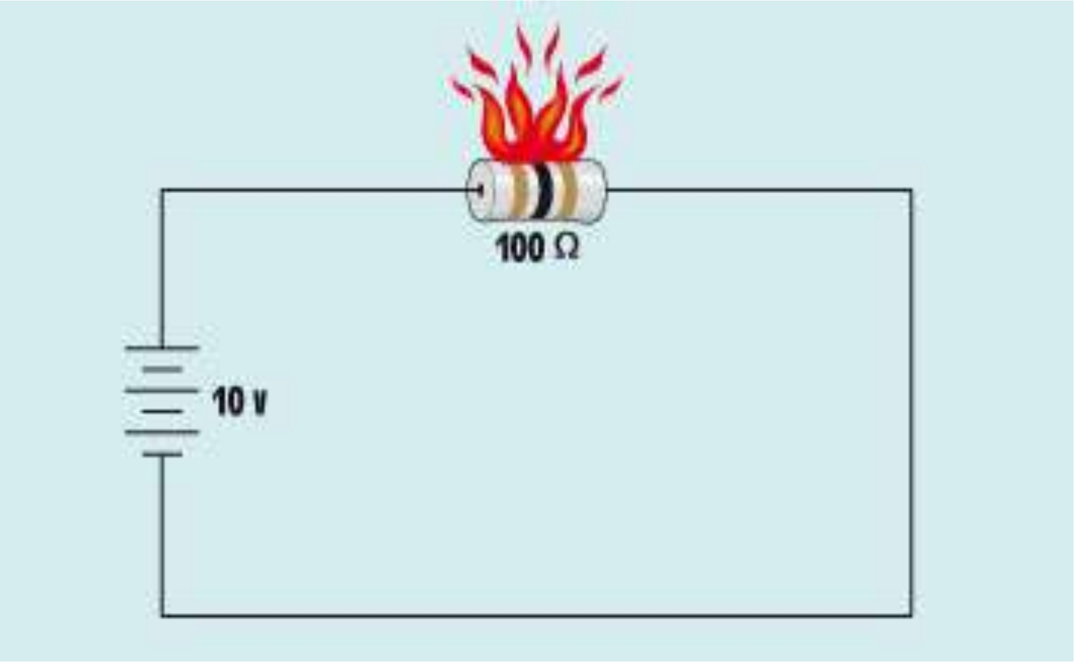
Exceeding the **power rating** causes damage to a resistor.



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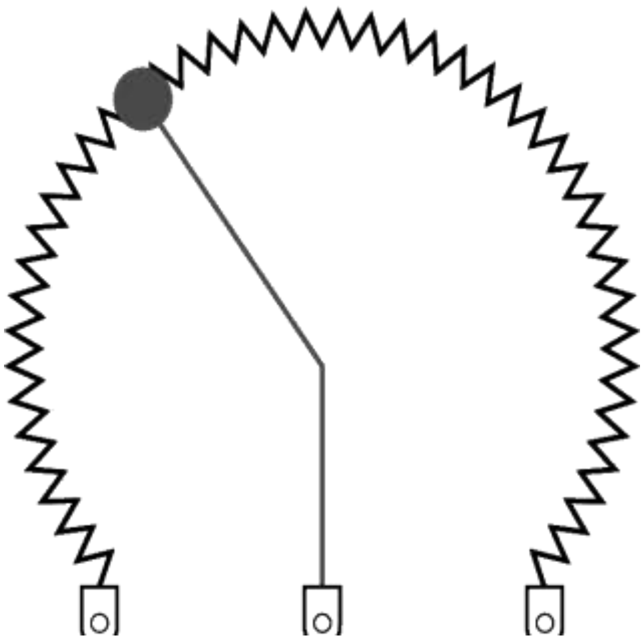
Variable resistors can change their value over a specific range. A **potentiometer** is a variable resistor with three terminals. A **rheostat** has only two terminals.



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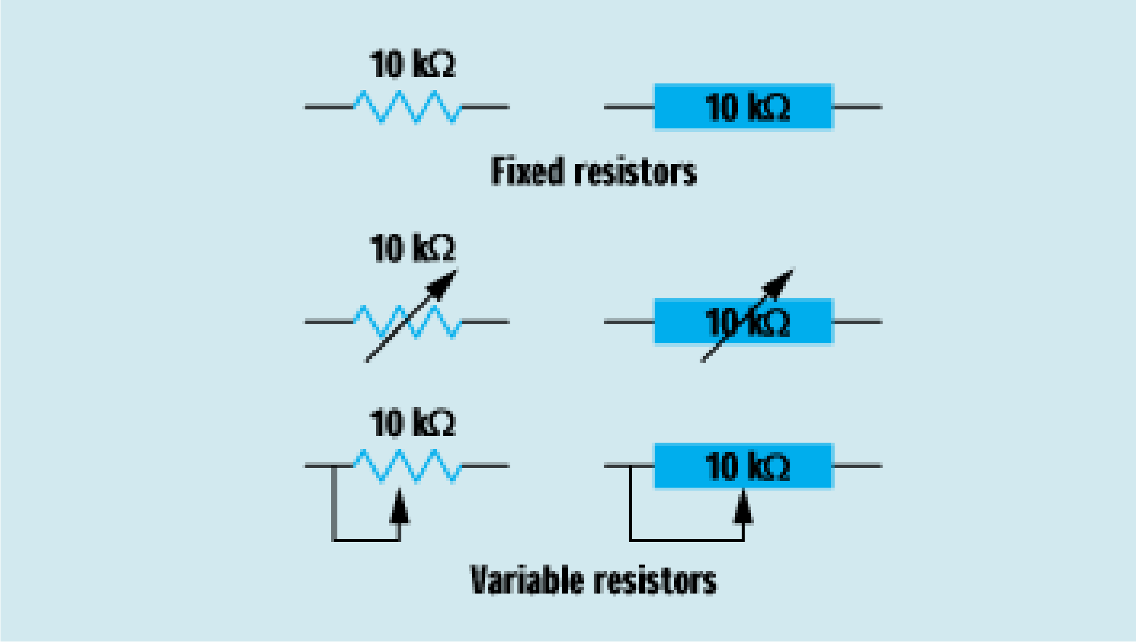
Schematic symbols are used to represent various types of fixed resistors.



Unit 5 Resistors

Review:

1. Resistors are used in two main applications: as voltage dividers and to limit the flow of current in a circuit.
2. The value of fixed resistors cannot be changed.
3. There are several types of fixed resistors such as composition carbon, metal film, and wire-wound.

Unit 5 Resistors

Review:

4. Carbon resistors change their resistance with age or if overheated.
5. Metal film resistors never change their value, but are more expensive than carbon resistors.
6. The advantage of wire-wound resistors is their high power ratings.

Unit 5 Resistors

Review:

7. Resistors often have bands of color to indicate their resistance value and tolerance.
8. Resistors are produced in standard values. The number of values between 0 and 100 Ω is determined by the tolerance.

Unit 5 Resistors

Review:

9. Variable resistors can change their value within the limit of their full value.
10. A potentiometer is a variable resistor used as a voltage divider.